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PRACTICAL WEEK 1

Task 1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

a) **Write a C code to create a process using fork() system call.**

b) **Read the user input as shell command.**

c) **Use a system call exec() to execute the shell command.**

d) **Use a wait() system call for waiting a parent process.**

e) **Display the PIDs of parent and child processes.**

1) Create a file

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano problem1.c
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$
```

Then press enter to open command

2) write a code

```
  GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    char command[100];

    // Read Linux command from user
    printf("Enter a Linux command: ");
    scanf("%s", command);

    // Create child process
    pid_t pid = fork();

    if (pid < 0)
    {
        printf("Process creation failed!\n");
        return 1;
    }
```

```c
    else if (pid == 0)
    {
        // Child Process
        printf("\nChild Process\n");
        printf("Child PID  : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());

        printf("\nExecuting command: %s\n\n", command);

        // Execute Linux command
        execlp(command, command, NULL);

        // Executes only if exec fails
        perror("exec failed");
        exit(1);
    }
    else
    {
        // Parent Process
        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID  : %d\n", pid);

        // Wait for child process
        wait(NULL);

        printf("\nChild process completed.\n");
    }

    return 0;
}
```

1) To save press control O
2) To come back to shell press control X
3) Then for compilation write a command

   gcc filename.c -o filename

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano problem1.c
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ gcc problem1.c -o problem1
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$
```

   Then press enter then to view a output write a command

   ./filename

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano problem1.c
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ gcc problem1.c -o problem1
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ ./problem1
Enter a Linux command:
```

   Then you will be asking to enter linux command to view the output

   In ENTER LINUX COMMAND: enter either "ls" command (or) "pwd" command to see the output of your code

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano problem1.c
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ gcc problem1.c -o problem1
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ ./problem1
Enter a Linux command: ls

Parent Process
Parent PID : 751
Child PID  : 752

Child Process
Child PID  : 752
Parent PID : 751

Executing command: ls

File2.txt  myprogram     practical3.1.c  practical3.2.c  problem1.c    skill2    skill3    skill3main2    skill3main2.txt  sum1    task2    teja    vishwa.c
linux      practical3.1  practical3.2    problem1        problem1.txt  skill2.c  skill3.c  skill3main2.c  skill4.1.c       sum1.c  task2.c  vishwa

Child process completed.
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$
```

Task-2: Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

#uname

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ uname
Linux
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$
```

#lscpu

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ lscpu
Architecture:             x86_64
  CPU op-mode(s):         32-bit, 64-bit
  Address sizes:          39 bits physical, 48 bits virtual
  Byte Order:             Little Endian
CPU(s):                   16
  On-line CPU(s) list:    0-15
Vendor ID:                GenuineIntel
  Model name:             13th Gen Intel(R) Core(TM) i7-13620H
    CPU family:           6
    Model:                186
    Thread(s) per core:   2
    Core(s) per socket:   8
    Socket(s):            1
    Stepping:             2
    BogoMIPS:             5836.80
    Flags:                fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_relia
                          ble nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3d
                          nowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsav
                          ec xgetbv1 xsaves avx_vnni vnmi umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_clear serialize ibt flush_l1d arch_capabilities
Virtualization features:
  Virtualization:         VT-x
  Hypervisor vendor:      Microsoft
  Virtualization type:    full
Caches (sum of all):
  L1d:                    384 KiB (8 instances)
  L1i:                    256 KiB (8 instances)
  L2:                     10 MiB (8 instances)
  L3:                     24 MiB (1 instance)
NUMA:
  NUMA node(s):           1
  NUMA node0 CPU(s):      0-15
Vulnerabilities:
  Gather data sampling:   Not affected
  Ghostwrite:             Not affected
  Indirect target selection: Not affected
  Itlb multihit:          Not affected
  L1tf:                   Not affected
  Mds:                    Not affected
  Meltdown:               Not affected
  Mmio stale data:        Not affected
  Old microcode:          Not affected
  Reg file data sampling: Mitigation; Clear Register File
  Retbleed:               Mitigation; Enhanced IBRS
  Spec rstack overflow:   Not affected
  Spec store bypass:      Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:             Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:             Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSB-eIBRS SW sequence; BHI BHI_DIS_S
  Srbds:                  Not affected
  Tsa:                    Not affected
  Tsx async abort:        Not affected
  Vmscape:                Not affected
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$
```

#ps

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ ps
    PID TTY          TIME CMD
    342 pts/0    00:00:00 bash
    790 pts/0    00:00:00 ps
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$
```

#top

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ top
top - 06:46:29 up 14 min,  1 user,  load average: 0.07, 0.04, 0.00
Tasks:  24 total,   1 running,  23 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 99.4 id,  0.6 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7755.6 total,   7099.1 free,    583.7 used,    224.8 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   7171.9 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   24132  15408  11608 S   0.0   0.2   0:00.70 systemd
      2 root      20   0    3180   2208   2072 S   0.0   0.0   0:00.01 init-systemd(Ub
      7 root      20   0    3180   2044   1940 S   0.0   0.0   0:00.00 init
     47 root      19  -1   50396  16852  15556 S   0.0   0.2   0:00.17 systemd-journal
     79 systemd+  20   0   22540  14628  12008 S   0.0   0.2   0:00.06 systemd-resolve
     86 root      20   0   35412  12336   9200 S   0.0   0.2   0:00.50 systemd-udevd
    108 root      20   0    2888   1948   1820 S   0.0   0.0   0:00.04 chronyd-starter
    109 root      20   0    4472   3156   2912 S   0.0   0.0   0:00.00 cron
    110 message+  20   0    8820   5380   4704 S   0.0   0.1   0:00.05 dbus-daemon
    130 root      20   0   44092  29708  14564 S   0.0   0.4   0:00.21 networkd-dispat
    161 root      20   0   18440   9428   8176 S   0.0   0.1   0:00.08 systemd-logind
    192 syslog    20   0  220548   5780   4492 S   0.0   0.1   0:00.07 rsyslogd
    222 _chrony   20   0   24248  11648   7092 S   0.0   0.1   0:00.09 chronyd
    242 _chrony   20   0   12092   2440   1800 S   0.0   0.0   0:00.01 chronyd
    268 root      20   0    5492   2664   2476 S   0.0   0.0   0:00.01 agetty
    272 root      20   0  123456  32620  18000 S   0.0   0.4   0:00.15 unattended-upgr
    340 root      20   0    3184   1112    980 S   0.0   0.0   0:00.00 SessionLeader
    341 root      20   0    3200   1248   1108 S   0.0   0.0   0:00.10 Relay(342)
    342 vishwa_+  20   0    6268   5532   3860 S   0.0   0.1   0:00.19 bash
    343 root      20   0    8648   5628   4784 S   0.0   0.1   0:00.01 login
    386 vishwa_+  20   0   22340  13592  10976 S   0.0   0.2   0:00.15 systemd
    390 vishwa_+  20   0   22916   4084   2080 S   0.0   0.1   0:00.00 (sd-pam)
    417 vishwa_+  20   0    6136   5416   3864 S   0.0   0.1   0:00.09 bash
    800 vishwa_+  20   0    8208   5664   3532 R   0.0   0.1   0:00.00 top
```